📝 Detailed Lecture Notes: Maximum Likelihood for the Normal Distribution

Video: Maximum Likelihood For the Normal Distribution, step-by-step!!!
Channel: StatQuest with Josh Starmer

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1. The Normal Distribution and Its Parameters

The **Normal Distribution** (or Gaussian Distribution) is a fundamental continuous distribution defined by two parameters:

1. **Mean (μ):** Determines the **location** or center of the curve [00:59].
   - A smaller μ shifts the curve left; a larger μ shifts it right.
2. **Standard Deviation (σ):** Determines the **width** or spread of the curve [01:16].
   - A larger σ makes the curve **shorter and wider**.
   - A smaller σ makes the curve **taller and narrower**.

The goal of MLE is to use observed data $(X_1, X_2, \ldots, X_n)$ to find the optimal μ and σ that maximize the likelihood function.

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2. The Likelihood Function Setup

A. The Normal Distribution PDF

The core of the MLE is the **Probability Density Function (PDF)** for the Normal Distribution (which is used as the likelihood function):

$$f(X) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}$$

B. The Combined Likelihood

For n independent data points $(X_1, X_2, \ldots, X_n)$, the overall **Likelihood Function** (L) is the **product** of the individual likelihoods for each data point [07:39]:

$$L(\mu, \sigma) = \prod_{i=1}^{n} \left[\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{X_i-\mu}{\sigma}\right)^2}\right]$$

C. The Log-Likelihood Function (Simplification) [09:08]

- Since the Likelihood Function is a complex product, we take its logarithm (Log-Likelihood) to simplify the calculus.
- The logarithm converts **multiplication into addition** and simplifies the **exponents** [09:42].
- This simplified Log-Likelihood function is what we actually differentiate:

$$\log L(\mu, \sigma) = -n \cdot \log(\sigma) - \frac{n}{2} \cdot \log(2\pi) - \frac{1}{2\sigma^2} \cdot \sum_{i=1}^{n}(X_i - \mu)^2$$

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3. Deriving the MLE for the Mean (μ)

To find the optimal μ, we treat σ as a constant and take the derivative of the Log-Likelihood with respect to μ [08:34].

1. **Derivative:** Calculate $\frac{\partial \log L}{\partial \mu}$.
2. **Set to Zero:** Set the derivative equal to zero to find the peak [17:04].
3. **Solve for μ:** Solving the equation yields the MLE for the mean:

$$\text{MLE for } \mu = \frac{\sum \mathbf{X_i}}{\mathbf{n}}$$

- **Result Interpretation:** The Maximum Likelihood Estimate for the population mean (μ) is simply the **average of the observed measurements** (the sample mean) [17:55]. This determines the center of the Normal Curve.

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4. Deriving the MLE for the Standard Deviation (σ)

To find the optimal σ, we treat μ (using the sample mean from step 3) as a constant and take the derivative of the Log-Likelihood with respect to σ [08:52].

1. **Derivative:** Calculate $\frac{\partial \log L}{\partial \sigma}$.
2. **Set to Zero:** Set the derivative equal to zero to find the peak [17:13].
3. **Solve for σ^2 (Variance):** Solving the equation yields the MLE for the variance (σ^2):

$$\text{MLE for } \sigma^{\mathbf{2}} = \frac{\sum(\mathbf{X_i} - \mu)^{\mathbf{2}}}{\mathbf{n}}$$

- **Result Interpretation:**
  - The MLE for the variance (σ^2) is the **sample variance**, calculated using n in the denominator (instead of $n - 1$) [18:32].
  - The **MLE for σ** (Standard Deviation) is the square root of this value. This estimate determines the optimal width of the Normal Curve that maximizes the likelihood of the observed data [18:41].

Conclusion: The mathematical proof confirms that our intuition is correct: the **sample mean** and the **sample standard deviation** (with an n in the denominator) are the most likely estimates for the true population parameters (μ and σ) [19:06].

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